

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title

AI-Assisted Risk Flagging for Emergency Department Triage in Resource-Constrained Settings

Student Name

Vishal Baboolal

Statement of the Problem

Emergency departments often deal with overcrowding, time pressure, and variation in triage decisions, especially when nurses must quickly identify patients who may be at higher risk (Tyler et al., 2024; Da'Costa et al., 2025). Although recent studies show that AI can support emergency triage, many are based on retrospective or single-site datasets and do not always show how the output would fit into real bedside decision-making (Agius et al., 2025; Tahernejad et al., 2024). This pilot will explore whether a simple AI-assisted risk-flagging tool, using vital signs and presenting complaint data, can help triage nurses identify patients who may need urgent review.

Previous Work

Tyler et al. (2024) reviewed how artificial intelligence has been used in hospital emergency department triage. Their review showed that AI tools may help with identifying high-risk patients earlier, improving prioritisation, and supporting resource allocation. However, one important point from the review is that many studies focus mainly on model performance, while less attention is given to how these tools would actually work in a busy clinical environment. This is relevant to the proposed pilot because a tool may perform well in testing but still needs to be practical and safe for triage nurses to use.

Da'Costa et al. (2025) also discussed the potential of AI-driven triage in emergency departments. The review highlighted possible benefits such as faster patient prioritisation, reduced waiting times, and better use of limited resources. At the same time, the authors raised concerns about data quality, bias, clinician trust, ethics, and implementation. These concerns are important because an AI triage tool should support the nurse's judgement rather than replace it.

Araouchi and Adda (2024) presented TriageIntelli, an AI-assisted multimodal triage system for health centres. Their study compared several machine learning models, including random forests, support vector machines, artificial neural networks, gradient boosting,

logistic regression, XGBoost, and a stacking model. The results suggested that machine learning can predict triage categories with promising accuracy. However, this also shows why accuracy alone is not enough, since a useful triage tool must also be understandable, usable, and tested in the setting where it will be applied.

Agius et al. (2025) used emergency department data from Mater Dei Hospital in Malta to predict Emergency Severity Index level, hospital admission, and admitting ward. Their XGBoost-based models showed that routinely collected clinical data can support decisions in the emergency department, especially when more clinical information is included. This paper is useful for the Mercer project because it shows how ED data can be used for more than just basic prediction. However, because the study was based on one hospital system, its findings would still need to be validated before being used in a different context.

Tahernejad et al. (2024) reviewed the use of artificial intelligence in triage during emergencies and disasters. The paper showed that AI may help with response time, resource management, real-time data transfer, and vital sign monitoring in high-pressure emergency settings. However, it also pointed out practical barriers such as staff training, trust in AI systems, equipment limitations, privacy concerns, and implementation challenges. This connects well with the Mercer scenario because a resource-constrained emergency department would need a tool that is simple, realistic, and designed around actual workflow limitations.

Gaps Identified

One major gap in the literature is limited external validation across different clinical settings. Many AI triage studies report promising results, but the models are often developed using retrospective or single-hospital data. This means the same model may not work as well in another hospital, especially in a resource-constrained Caribbean emergency department. A 12-week Mercer pilot can begin to address this by testing a simple risk-flagging approach on local or Mercer-style triage data before considering wider use.

A second gap is the difference between model performance and real clinical usefulness. Several studies report accuracy, AUC, or model comparisons, but they do not always explain how a nurse would use the output during triage. In practice, the tool must be clear, quick to interpret, and safe to use. This pilot can address that by designing the output as a simple “urgent review” risk flag, along with the abnormal findings that triggered the flag, rather than trying to replace the triage nurse or automatically assign a final triage category.

Proposed Solution

The proposed solution is a 12-week pilot of a simple AI-assisted risk-flagging tool for emergency department triage. The tool would use information that is normally available during triage, such as vital signs, age, presenting complaint, and basic clinical observations. Its purpose would be to highlight patients who may need urgent review, not to replace the triage nurse or make final clinical decisions.

The pilot would begin by deciding which input variables and safety rules should be included. The data would then be cleaned and checked for missing, unrealistic, or inconsistent values. A simple baseline model could be developed and compared with basic rule-based clinical thresholds. The output would be kept easy to understand by showing the patient's risk flag and the main abnormal findings that contributed to it.

The success of the pilot would be judged using practical measures such as how well the tool identifies high-risk patients, how many false alerts it creates, how it handles missing data, and whether triage staff can understand the output. At the end of the 12 weeks, the expected deliverables would be a small prototype, a short evaluation report, and a recommendation on whether the idea should move forward for further testing. This keeps the project realistic, clinically relevant, and suitable for the timeframe.

References

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